

TEST-03-AEGIS-HOME-VALIDATION

TEST-03: AEGIS HOME VALIDATION PROTOCOL

Carrington Storm Motors — Project AEGIS

Whole-House GIC Susceptibility • Individual Product Validation • Installation QA • Durability • EMC

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VALIDATION PHILOSOPHY

Aegis Home is an integrated system of 18 products — not a collection of individually validated components. The system-level validation protocol verifies that the products work together to achieve the target 94% GIC reduction. Individual product validation ensures each component meets its specification. Installation QA ensures the system is built correctly. Durability testing ensures it stays correct for decades.

PHASE 1: INDIVIDUAL PRODUCT VALIDATION (18 Products)

P1.1 Product Validation Matrix

#	Product	Key Metrics	Test Standard	Pass/Fail	Test Duration	Cost
1	Aegis-C Wall Panel (1.2×2.4 m)	SE >148 dB 1 kHz–10 GHz; UL 94 V-0; E84 Flame Spread <25	TEST-02 MXene + UL 94/723	All metrics, all panels	4 months	\$180K
2	BFRP Rebar (8–32 mm)	Tensile >1,100 MPa; dielectric >15 kV/mm; bond strength with geopolymer >10 MPa	ASTM D7205, D7914, D149	All metrics, all sizes	4 months	\$75K
3	Dielectric Door Hinge	Load capacity >200 kg; 100K cycle endurance; DC resistance hinge-to-panel >10 ⁸ Ω	Custom test fixture	All metrics	2 months	\$28K
4	Inductive-Proof Wiring (PMMA POF)	Gigabit optical data at 100 m; MXene SE >85 dB on fiber bundle; bend radius <30 mm without loss	TIA/EIA-455 (FOTP); TEST-02 MXene	All metrics	3 months	\$45K
5	Conductive-Safe Pipe (GFRP)	Pressure >150 psi sustained; dielectric joint >10 ⁸ Ω; UV resistance per ASTM G154 SE >80 dB at 2.45	ASTM D1598 (pressure); TEST-02 UV	All metrics	3 months	\$38K

6	Safe-Wave Microwave	GHz; cooking performance equivalent to 800W conventional	FCC Part 18 (ISM); UL 923	All metrics	3 months	\$42K
7	Stellar-Chill Refrigerator	Electrocaloric cooling $\Delta T > 15^{\circ}\text{C}$; COP > 3.0 ; zero compressor/vibration	Custom calorimeter test	All metrics	3 months	\$35K
8	Carrington-Proof Oven	Temperature uniformity $\pm 10^{\circ}\text{C}$; energy consumption < 1.5 kWh for standard bake cycle	UL 858 modified	All metrics	3 months	\$28K
9	Inflatable Ceramic Water Heater	Heat time < 30 min to 60°C (40 gal); dielectric isolation $> 10^8 \Omega$ tank-to-water	UL 174 modified; custom elect. safety	All metrics	3 months	\$32K
10	Safe Cookware Set	Surface resistivity $> 10^{12} \Omega/\text{sq}$; YInMn Blue thermal stability to 500°C ; food safety (FDA 21 CFR)	Custom; FDA migration testing	All metrics	2 months	\$18K
11	EMF Detector	Sensitivity $0.1 \mu\text{T}$ - 10 T; 0 - 1 kHz; piezo-powered (no battery); LoRa mesh	Custom calibration in TEM cell	All metrics	2 months	\$22K
12	Stealth-Lift Garage Door	BFRP frame load > 500 kg; MXene mesh SE > 80 dB at door seams; 10K cycle endurance	Custom structural + EMI	All metrics	2 months	\$35K
13	Conductive-Proof Spray Cans	Applied SE > 80 dB at $45 \mu\text{m}$ dry film thickness; adhesion 5B per ASTM D3359	TEST-02 MXene + adhesion	All metrics	1 month	\$8K
14	Dielectric Rope	Breaking strength > 20 kN (12 mm dia); dielectric > 20 kV/mm; UV resistance	ASTM D4268 (rope); TEST-02 dielectric/UV	All metrics	2 months	\$15K
15	Non-Conductive Motor (2 HP)	Efficiency $> 85\%$; torque ripple $< 5\%$; motor-to-chassis resistance $> 10^8 \Omega$	IEEE 112 (motor efficiency); custom GIC	All metrics	4 months	\$48K
16	Conductive-Proof Tool Set	PEEK CF40 strength > 220 MPa; ceramic blade hardness $> \text{HV} 15$ GPa; 100% non-sparking	ASTM D638 (tensile); ASTM E92 (Vickers)	All metrics	1 month	\$12K
17	Aerogel Thermal Breaks	$\lambda < 0.015$ W/m·K; compressive > 0.8 MPa; R-value > 10 per inch	ASTM C518; ASTM C165	All metrics	2 months	\$15K
18	Safe Stove Ventilation	Airflow > 200 CFM; dielectric fan motor; duct GIC isolation $> 10^8 \Omega$	Custom airflow + EMI	All metrics	2 months	\$18K

Phase 1 Total Cost: \$689K; Total Duration: 12 months (parallelized across products)

PHASE 2: WHOLE-HOUSE GIC SUSCEPTIBILITY TESTING

P2.1 Test Objectives

Verify that a complete Aegis Home (all 18 products installed) achieves >94% reduction in induced ground current compared to an unhardened control home of identical construction.

P2.2 Test Setup

Test Site: CSM El Segundo outdoor test pad (or Aegis Home prototype per M2.4)
Test Homes: - **DUT (Device Under Test):** Full-scale 1,800 sq ft Aegis Home prototype, all 18 products installed per CSMFAB specifications, 48 GIC sensors embedded in foundation, walls, wiring, plumbing, and appliances - **Control:** Identical floor plan constructed with conventional materials (steel rebar, copper wiring, metal plumbing, standard appliances), also with 48 sensors, located >50 m away (to prevent EM coupling between homes)

P2.3 GIC Simulation Method

Option A — Current Injection (Preferred): Inject controlled current (0.1–50 A RMS, 0.001–1 Hz) into ground electrodes surrounding each home to simulate GIC. Measure induced currents in structural, wiring, and plumbing loops. Compare DUT vs. Control.

Option B — TEM Cell Scaling: Construct 1:10 scale models of DUT and Control homes; place in TEM cell; apply scaled EM fields. Extrapolate to full scale using EM scaling laws. Less accurate than Option A but useful for early-stage screening.

Option C — Geomagnetic Storm Correlation: Wait for actual geomagnetic storm ($K_p \geq 5$) and compare real-time GIC measurements between DUT and Control during the event. Uncontrollable timing; used as supplementary validation.

Selected: Option A (current injection) for primary validation; Option C for ongoing monitoring during pilot.

P2.4 Instrumentation (48 Sensors per Home)

Sensor Type	Quantity	Location	Measurement	Sampling
Rogowski Coil (flexible CT)	12	Foundation rebar loops, wall panel edges, roof ridge	AC current (0.01–100 A, 0.1 Hz–1 kHz)	1 Hz
Voltage Differential Probe	8	Between plumbing segments, across dielectric joints, panel-to-panel	AC/DC voltage (0.1 mV–600 V)	1 Hz
Surface Electric Field Mill	4	N, S, E, W exterior walls at 1 m height	E-field (1 V/m–50 kV/m)	1 Hz
Magnetometer (Fluxgate)	2	Interior center of home at floor and ceiling	B-field (0.1 nT–100 μ T, DC–1 kHz)	10 Hz
Fiber Optic Strain Gauge (FBG)	10	Foundation corners, wall midpoints, roof trusses	Microstrain (structural response to GIC-induced forces)	100 Hz
Temperature (Thermocouple)	6	Panel interior surfaces, plumbing, appliance housings	Temperature (–40 to +125°C)	0.017 Hz (1/min)
Humidity Sensor	2	Interior ambient, crawlspace	RH%	0.017 Hz
Current Transformer (utility feed)	2	Main service entrance (DUT — dielectric isolated vs. Control —	Service current, neutral-to-ground	60 Hz (power quality)

Reference Ground Electrode	2	conventional bonded neutral)	current	Earth potential reference	1 Hz
		Remote earth reference, 100 m from test site			

P2.5 Test Procedure — Current Injection

1. Install ground electrode array: 4 electrodes at 10 m spacing in square pattern around each home
2. Connect controlled current source (programmable AC power supply + step-up transformer, 0-50 A, 0.001-1 Hz)
3. Baseline measurement: All sensors recording, zero injected current, 24 hours (characterize ambient GIC from natural geomagnetic activity)
4. Injection sweep: Inject current at 11 frequencies (0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 5, 10, 50, 100 Hz), at 5 amplitudes per frequency (0.5, 1, 5, 10, 50 A)
5. For each frequency/amplitude combination: record all sensors for 10 minutes (≥10 cycles at lowest frequency)
6. Between sweeps: 5-minute zero-current interval to allow transient decay
7. Total test duration: ~12 hours per home

P2.6 Data Analysis — GIC Reduction Calculation

For each sensor location and each injected current frequency/amplitude: 1. Calculate transfer function: $H(f) = I_{\text{induced}}(f) / I_{\text{injected}}(f)$ for DUT and Control 2. GIC Reduction Ratio = $H_{\text{Control}}(f) / H_{\text{DUT}}(f)$ (unitless ratio) 3. Report as percent reduction: $\text{GIC_Reduction}(\%) = (1 - 1/\text{Ratio}) \times 100$ 4. Aggregate across all sensor locations to compute mean reduction with 95% confidence interval 5. Primary metric: Mean GIC reduction across all 12 Rogowski coil locations

P2.7 Pass/Fail Criteria

Criterion	Threshold	Measurement
PRIMARY: Whole-Home GIC Reduction	>94% reduction (mean across all Rogowski coil locations at 0.01-1 Hz range — the geomagnetic band)	Current injection test
PRIMARY: Zero Safety Events	No measured current >5 mA through any human-touchable surface (door handles, faucets, appliance cases, light switches) at maximum injected current (50 A)	Voltage differential probes at touch points; calculate body current via IEC 60479 body impedance model
SECONDARY: Panel SE	Every Aegis-C wall panel maintains >140 dB SE when installed (vs. >148 dB at individual panel level — 8 dB degradation acceptable for joint losses)	Portable SE test fixture at 10 panel locations
SECONDARY: Wiring GIC Isolation	PMMA POF wiring carries zero measurable induced current (<1 µA) at any injected current level	Rogowski coils on POF bundles
SECONDARY: Plumbing GIC Isolation	Current across any dielectric pipe joint <10 µA at maximum injected current	Rogowski coils at all 16 dielectric joints
SECONDARY: Structural Integrity	Zero structural damage (cracking, delamination, displacement >1 mm) after maximum current injection	Pre/post visual inspection; FBG strain gauge data (Δmicrostrain <100 at any location)

PHASE 3: INSTALLATION QUALITY ASSURANCE

P3.1 QA Checkpoints During Construction

Construction Phase	QA Checkpoint	Instrument/Method	Acceptance Criteria
	BFRP rebar placement per		

Foundation Pour	Rebar location placement per drawing; geopolymer mix verified (slump, density); no steel rebar present (metal detector)	Visual inspection; slump cone; nuclear density gauge; handheld metal detector	Rebar spacing ± 10 mm; slump 150–200 mm; density 2,200–2,400 kg/m ³ ; zero metal detected
Wall Panel Installation	Panel orientation correct (exterior face outward); MXene layer continuity at joints (no gaps >2 mm); gasket compression (torque wrench on bolts to 15 N·m)	Visual; feeler gauge at panel joints; torque wrench	All panels oriented correctly; gaps <2 mm everywhere; all bolts 15 ± 2 N·m
PMMA POF Wiring	No copper wiring present (metal detector sweep of all walls); POF bend radius >30 mm everywhere; OTDR test on each fiber (insertion loss <0.3 dB/connector)	Metal detector; bend radius gauge; OTDR (EXFO or Viavi)	Zero metal detected; all bends >30 mm; all fibers <0.3 dB loss per connector, <3 dB total link loss
GFRP Plumbing	Dielectric joint gap >10 mm (verify no metal bridging); pressure test at 150 psi for 24 hours (zero pressure drop)	Feeler gauge; pressure gauge with chart recorder	All gaps >10 mm; pressure drop <1 psi over 24 hours
Appliance Installation	Ground resistance measurement (appliance chassis to earth > $10^8 \Omega$); functional test (all modes operational)	Megohmmeter (1,000 V test voltage); functional checklist	All $R > 10^8 \Omega$; all functions operational
Final System Integration	Whole-home GIC susceptibility test (current injection at 5 A, 1 Hz — production screening test, not full characterization)	Portable GIC injection kit (5 A max); 6 Rogowski coils at key locations	Zero measurable GIC in wiring or plumbing; panel SE within 5 dB of individual panel spec

P3.2 Installation Defect Categories

Defect Category	Definition	Action
CRITICAL	Creates conductive path where dielectric isolation is required; compromises GIC reduction target	Immediate rework; re-inspection before construction proceeds; root cause analysis
MAJOR	Reduces EMI shielding effectiveness by >3 dB at any frequency; creates safety hazard per NEC	Rework within 24 hours; re-inspection
MINOR	Cosmetic issue; dimensional tolerance out of spec but does not affect GIC/EMI/safety performance	Document in punch list; resolve before homeowner occupancy
OBSERVATION	Potential improvement identified; does not violate specification	Document for continuous improvement database

PHASE 4: LONG-TERM DURABILITY — ACCELERATED AGING

P4.1 Environmental Exposure Testing

Exposure	Method	Duration	Post-Exposure Verification
UV + Moisture (exterior panels)	ASTM G154 Cycle 1 (UVA-340, 8h UV 60°C / 4h condensation 50°C)	2,000 hours (~5 years Florida equivalent)	SE within 5 dB of initial; YInMn Blue $\Delta E < 3$ (colorfastness); no delamination
Thermal Cycling (panels +	-20°C to +60°C 5°C/min	365 cycles (1 year diurnal	SE within 3 dB; no cracking

Thermal Cycling (panels + structural)	20 °C to 100 °C, 5 °C/min, 1 hour dwell	300 cycles (1 year thermal equivalent)	>0.1 mm; structural FBG Δstrain <50 μstrain from baseline Mass loss <5%; dynamic modulus >80% of initial; resistivity >10 ⁷ Ω·m Functional test pass; no mold/mildew; SE within 3 dB; electrical safety pass SE within 5 dB; no corrosion on any surface (no metal to corrode — verify no unexpected degradation of MXene, BFRP, or ceramic components) No water penetration through panel joints; interior RH increase <5%
Freeze-Thaw (geopolymer foundation)	ASTM C666 Procedure A (rapid freezing and thawing in water)	300 cycles	
High Humidity (interior products)	40°C / 93% RH	1,000 hours	
Salt Spray (coastal homes)	ASTM B117	500 hours	
Wind-Driven Rain	ASTM E331 (static pressure 15% of design wind pressure)	15 minutes per wall	

P4.2 Mechanical Endurance

Product	Test	Cycles/Load	Acceptance
Dielectric Door Hinge	Open/close cycle — 100K cycles at 50 kg door weight	100,000	No play >1 mm; DC resistance hinge-to-panel >10 ⁸ Ω
Stealth-Lift Garage Door	Open/close cycle	10,000	No SE degradation >3 dB at seams; motor current within 10% of initial
Window Seals (aerogel thermal breaks)	Compression cycling (ASTM C864 for sealants)	5,000 cycles, ±2 mm displacement	Air leakage <0.3 cfm/ft² at 75 Pa per ASTM E283

PHASE 5: EMC COMPLIANCE — WHOLE-HOME

P5.1 Radiated Emissions

Test	Standard	Method	Limit
Radiated emissions (30 MHz–1 GHz)	FCC Part 15, Class B (residential)	CISPR 16-1-4, 3 m measurement distance	40 dBμV/m at 3 m (30–88 MHz), 43.5 dBμV/m (88–216 MHz), 46 dBμV/m (216–960 MHz), 54 dBμV/m (above 960 MHz)
Conducted emissions (150 kHz–30 MHz)	FCC Part 15, Class B	LISN on mains	66–56 dBμV (quasi-peak), 56–46 dBμV (average), decreasing with frequency

Expected Result: Aegis Home with dielectric wiring and non-metallic construction is inherently quieter than conventional home. All products expected to pass with >10 dB margin.

P5.2 Immunity

Test	Standard	Level	Performance Criterion
IEC 61000-4-2 (ESD)	±8 kV contact, ±15 kV air	Level 4 (highest)	Criterion A (normal performance during and after test)
IEC 61000-4-3 (Radiated RF)	80 MHz–6 GHz, 10 V/m	Level 3 (residential)	Criterion A
IEC 61000-4-4 (EFT/Burst)	±2 kV on power lines	Level 3	Criterion B (temporary)

IEC 61000-4-5 (Surge)	±2 kV line-to-earth	Level 3	degradation, self-recoverable)
IEC 61000-4-6 (Conducted RF)	150 kHz-80 MHz, 10 V	Level 3	Criterion B
IEC 61000-4-8 (Power Frequency Magnetic Field)	100 A/m continuous, 1,000 A/m for 3 s (special Carrington test level — standard only requires 30 A/m and 300 A/m)	Custom “Level X”	Criterion A
			Criterion A at 1,000 A/m (confirms GIC immunity)

FIELD TRIAL PROTOCOL (REFERENCE TO ROAD-06)

The Aegis Home field trial (50 homes, 12 months) is defined in ROAD-06 (Pilot 1). Key validation data collected during field trial:

- Continuous GIC Monitoring:** 48 sensors per home, 1 Hz continuous, correlated with NOAA SWPC geomagnetic data
- Energy Performance:** HERS Index rating at 12 months; compare to control homes
- Occupant Satisfaction:** Validated POE survey at 3, 6, 12 months
- Maintenance Events:** All maintenance logged; categorize by product, severity
- Health Outcomes (Optional):** HRV, sleep quality, cortisol (pre/post comparison) — IRB-approved study

Field Trial Success Criteria (from ROAD-06)

Criterion	Threshold
GIC Reduction	>94% reduction vs. matched control homes
Zero Electrical Safety Events	Zero shock/fire incidents attributable to electrical system at 12 months
Occupant Satisfaction	>90% “satisfied” or “very satisfied”
Durability	<2% of components requiring repair/replacement at 12 months
Insurance Premium Impact	Documented premium reduction of >5%

PERFORMANCE METRICS — FINAL SYSTEM SPECIFICATION

Metric	Target	Verification Method
Whole-Home GIC Reduction	>94% (mean across all current-carrying paths at 0.01-1 Hz)	Phase 2 current injection test
Aegis-C Panel SE (installed)	>140 dB (all panels, all locations, 1 kHz-10 GHz)	Portable SE test fixture
Wiring GIC Current	<1 µA (any PMMA POF bundle, maximum injection)	Rogowski coil
Plumbing GIC Current	<10 µA (any dielectric joint, maximum injection)	Rogowski coil
Structural GIC Current	<100 µA (any foundation rebar loop, maximum injection)	Rogowski coil
Touch-Point Body Current	<5 mA through 1 kΩ body impedance model (any human-touchable surface)	Voltage measurement + IEC 60479 model
Enclosure SE (appliances)	>40 dB at 2.45 GHz (microwave); >30 dB at 100 MHz-1 GHz (electronics)	Portable SE test fixture
DC Isolation (any conductive path)	>10 ⁸ Ω to earth or between any two points not intended to be connected	Megohmmeter at 1,000 V
Energy Performance	HERS Index <45	RESNET HERS rater
Durability	All components functional within	Phases 4a, 4b

“Aegis Home is the flagship. When a homeowner asks ‘will this house really protect my family from the next Carrington Event?’ the answer is not in the brochure — it is in the 48-sensor GIC injection test, the 2,000-hour UV chamber, the 300-cycle freeze-thaw test, and the 12-month field trial data from 50 homes. This validation protocol converts hope into evidence.” — Operative Corvus Nyx (AEGIS HOME)

Document generated 2026-07-15. Phase 1 (individual product validation) is in progress. Phase 2 (whole-home GIC test) requires completion of Aegis Home prototype (M2.4, Q1 2027). Phase 4 (accelerated aging) initiates upon prototype completion and runs concurrently with pilot. All test methods are NIST-traceable. Data published per open-data policy defined in TEST-01.